Lab Practice II

Mtech CSE Semester II 2025 -2026

List of Assignments

Course description : This laboratory course provides hands-on exposure to advanced deep learning models, data analytics, visualization tools, and end-to-end MLOps practices. Students implement real-world problems involving image processing, NLP, recommender systems, business intelligence dashboards, and cloud-deployed machine learning pipelines, aligning with industry and research expectations.

Course Objectives

● To implement advanced deep learning models for vision and language tasks ● To apply data preprocessing, modeling, and visualization techniques on real datasets ● To design and deploy scalable ML systems using MLOps best practices ● To build reproducible and automated ML pipelines using industry tools

Course Outcomes:

● CO1: Build end-to-end AI-based systems for real-world problem domains. ● CO2: Apply deep learning techniques to diverse datasets including images, text, signals, and sensor data.
● CO3: Create analytical dashboards for data interpretation and decision support. ● CO4: Deploy and manage machine learning models using MLOps best practices.

Students have to choose any four problem statement for implementation

Problem Statement 1: Intelligent Image Classification and Visual Analytics Platform

Description:
Develop an image classification platform using deep neural networks to classify images from benchmark datasets. Analyze model performance metrics and dataset characteristics using visual analytics tools. Design and implement an automated MLOps workflow to manage experiments, model packaging, and continuous deployment.

Core Components:

- Deep Learning: Deep MLP / CNN for image classification
- Data Visualization: Performance analysis dashboards (accuracy, loss, confusion matrix)
- MLOps: CI/CD pipeline, Docker-based deployment, MLflow experiment tracking

Expected Deliverables:

Optimized classification model
Visual analytics dashboard
Automated CI/CD workflow

Problem Statement 2: Smart Plant Health Monitoring System

Description:
Design and develop an intelligent plant health monitoring system that detects crop diseases from leaf images using deep learning. Perform data preprocessing and exploratory analysis on the dataset and present disease occurrence trends using interactive dashboards. Implement an MLOps pipeline to version datasets, track experiments, and deploy the trained model as a cloud-accessible API.

Core Components:

● Deep Learning: CNN-based multi-class plant disease classification
● Data Visualization: EDA and disease trend dashboards using Tableau / Power BI ● MLOps: Data versioning (Git + DVC), experiment tracking (MLflow), API deployment

Expected Deliverables:

● Trained CNN model and evaluation report
● Interactive visualization dashboard
● Deployed prediction API and reproducible pipeline

Problem Statement 3: Sentiment Intelligence System for Product Reviews

Description:
Build a sentiment analysis system that classifies textual reviews using LSTM-based deep learning models. Perform data preparation and statistical analysis to understand sentiment distribution and trends. Deploy the system using MLOps best practices with model monitoring and reproducible experimentation.

Core Components:

● Deep Learning: LSTM-based sentiment classification
● Data Visualization: Sentiment trends and distribution dashboards
● MLOps: Experiment tracking, model versioning, REST API deployment **Expected**

Deliverables:

- Trained sentiment analysis model
- Interactive sentiment visualization dashboard
- Deployed inference service with tracked experiments

Problem Statement 4: Smart Recommendation and Insight Dashboard System

Description:
Design a personalized recommendation system using deep learning techniques to predict user preferences. Perform data aggregation and exploratory analysis to generate insights on user behavior and system performance. Implement an end-to-end MLOps pipeline to automate model training, deployment, and lifecycle management.

Core Components:

- Deep Learning: Deep learning–based recommendation model
- Data Visualization: User behavior and recommendation insights dashboard
- MLOps: Data versioning, automated training, model deployment pipeline

Expected Deliverables:

- Recommendation engine with evaluation metrics
- Interactive BI dashboard
- End-to-end automated ML pipeline

Problem Statement 4: Smart Rainfall Prediction and Decision Support System

Description:
Develop a rainfall prediction system using machine learning/deep learning models trained on historical weather data. Perform temporal and spatial data analysis to visualize rainfall trends. Implement a full MLOps architecture to enable reproducible training, experiment tracking, automated deployment, and continuous integration.

Core Components:

- Deep Learning: Rainfall prediction model
- Data Visualization: Time-series and geospatial rainfall dashboards
- MLOps: Git + DVC, MLflow, FastAPI, Docker, CI/CD

Expected Deliverables:

- Trained rainfall prediction model
- Cloud-based analytics dashboard
- Fully automated MLOps pipeline

Problem Statement 5 : AI-Driven Home-Based Parkinson's Rehabilitation Monitoring System

Description:

Design and develop an AI-driven home-based rehabilitation monitoring system for individuals with Parkinson's disease using wearable sensor data and/or video recordings captured in a non-clinical environment. The system should automatically assess motor performance during prescribed rehabilitation exercises, track patient adherence and progress over time, and provide actionable insights to clinicians or caregivers. Deep learning models will be used to quantify motor symptoms such as tremor, bradykinesia, gait irregularities, and postural instability. Data visualization techniques will be applied to effectively communicate rehabilitation outcomes. An end-to-end MLOps pipeline will be implemented to ensure reproducibility, scalability, and continuous model improvement.

Core Components:

- Deep Learning: CNN/LSTM or pose-estimation–based models for motor symptom evaluation and rehabilitation progress tracking
- Data Visualization: Home-based rehabilitation dashboards showing exercise adherence, symptom severity trends, and improvement metrics
- MLOps: Git + DVC for data versioning, MLflow for experiment tracking, FastAPI/Docker deployment, CI/CD-enabled updates

Expected Deliverables:

- **Trained home-based rehabilitation assessment model**
- **Interactive dashboards for patient progress and therapy effectiveness**
- **Deployed and reproducible MLOps pipeline suitable for remote monitoring**

Problem Statement 6: Parkinson's Disease Detection Using Brain Imaging Data

Description:

Design and implement an intelligent system for the detection of Parkinson's disease using brain imaging data (MRI or DaTscan). Develop a deep learning model to differentiate Parkinson's patients from healthy controls by learning disease-specific neuroimaging patterns. Perform image preprocessing and exploratory visual analysis to study structural

or functional differences. Implement an end-to-end MLOps pipeline to ensure reproducibility, experiment tracking, and deployment of the trained model for inference.

Core Components:

- **Deep Learning: CNN / transfer-learning models for Parkinson's detection from neuroimaging data**
- **Data Visualization: Brain slice visualization, class-wise feature analysis, and performance dashboards**
- **MLOps: Git + DVC for data versioning, MLflow for experiment tracking, Dockerized REST API deployment**

Expected Deliverables:

- **Trained Parkinson's detection model with evaluation metrics**
- **Visual analytics dashboard highlighting imaging patterns and model performance**
- **Deployed inference service with reproducible MLOps pipeline**

Problem Statement 7: MRI-Based Epilepsy Detection System

Description:

Design and implement an automated epilepsy detection system using structural MRI brain images. Develop a deep learning model to distinguish epileptic patients from healthy controls by learning spatial brain patterns associated with epilepsy. Perform image preprocessing and exploratory visual analysis to highlight structural variations and model performance. Implement an end-to-end MLOps pipeline to manage datasets, track experiments, and deploy the trained model for inference.

Core Components:

- Deep Learning: CNN-based classification of MRI brain images (e.g., T1-weighted MRI from public epilepsy datasets)
- Data Visualization: MRI slice visualization, class-wise feature analysis, and performance dashboards
- MLOps: Data versioning (Git + DVC), experiment tracking (MLflow), Dockerized model deployment via REST API

Expected Deliverables:

- Trained MRI classification model with evaluation metrics
- Visual analytics dashboard for MRI features and results
- Deployed inference service with reproducible pipeline

Problem Statement 8: EEG-Based Epileptic Seizure Detection System

Description:
Design and implement an intelligent system to detect epileptic seizures from EEG signals using deep learning techniques. Perform signal preprocessing and exploratory analysis to visualize EEG patterns across seizure and non-seizure states. Develop an end-to-end MLOps pipeline to ensure reproducible experiments, model versioning, and deployment of the seizure detection model as a real-time inference service.

Core Components:

- Deep Learning: CNN/LSTM-based seizure detection using EEG datasets (e.g., Bonn EEG Dataset, CHB-MIT)
- Data Visualization: Time-series EEG visualization, feature distributions, and performance dashboards
- MLOps: Data versioning (Git + DVC), experiment tracking (MLflow), FastAPI-based model deployment

Expected Deliverables:

- Trained seizure detection model with evaluation metrics
- EEG signal analysis and visualization dashboard
- Deployed inference API with reproducible training pipeline

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	3	3	-	-	-	-	2	-	-	3	2
CO2	3	3	2	-	3	-	-	-	-	-	-	1	3	2
CO3	2	2	-	2	3	-	-	-	-	3	2	-	2	3
CO4	2	3	-	3	3	-	-	-	-	2	3	2	3	4

Justification for CO–PO Mapping

CO1: Build end-to-end AI-based systems

- PO1 (High – 3): Requires strong foundational knowledge in AI, ML, and computing systems.
- PO2 (High – 3): Involves problem analysis and solution formulation for real-world

domains.

- PO3 (Moderate – 2): Includes investigation through data collection, experimentation, and evaluation.
- PO4 (High – 3): Focuses on designing complete AI systems.
- PO5 (High – 3): Extensive use of modern tools, frameworks, and platforms.
- PO10 (Moderate – 2): Requires documentation and presentation of system designs and results.

CO2: Apply deep learning to diverse datasets

● PO1 (High – 3): Strong application of deep learning theory and algorithms.
● PO2 (High – 3): Analysis of dataset characteristics and suitable DL models.
● PO3 (Moderate – 2): Experimental evaluation and interpretation of results.
● PO5 (High – 3): Use of modern DL frameworks and computational tools. ● PO12 (Low – 1): Encourages continuous learning due to rapidly evolving DL techniques.

CO3: Create analytical dashboards

● PO1 (Moderate – 2): Applies computing and data analytics fundamentals. ● PO2 (Moderate – 2): Involves identifying data insights for decision support. ● PO4 (Moderate – 2): Design of dashboards and visualization systems. ● PO5 (High – 3): Uses tools for visualization, analytics, and reporting. ● PO10 (High – 3): Strong emphasis on effective communication of insights. ● PO11 (Moderate – 2): Incorporates project planning and management aspects.

CO4: Deploy and manage ML models using MLOps

● PO1 (Moderate – 2): Applies ML and software engineering fundamentals. PO2 (High – 3): Analysis of deployment, monitoring, and scalability challenges. PO4 (High – 3): Design of ML deployment pipelines and systems.
PO5 (High – 3): Extensive use of DevOps/MLOps tools and platforms. PO10 (Moderate – 2): Requires documentation and reporting of pipelines. PO11 (High – 3): Strong alignment with project and operations management. PO12 (Moderate – 2): Emphasizes lifelong learning due to evolving MLOps practices.

Part I: Continuous Assessment Marks [50]

Marking Scheme for

Criteria for Continuous Assessment (CA)

Academic Year - 2025-26 Semester II

Subject: Laboratory Practice - II

Project Based Assessment 1 Assignments

PREPARED BY

Dr. Reena Pagare

Assessmen
Problem Und Formulat Learning, DI
Algorithm Im Result A Evaluation: met
Minor pr problem stat on Deep l DMV+ implementat - Repo Docum (Structured code snippets conclu

Active partici discussions, Q consistent at

Class Interaction / Attendence